OSSP PRACTICAL 1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

File Creation, Compilation and Execution: The C program is created and compiled successfully. The program is executed, and the ls command is given as input to the child process. The child process executes the command using execlp().

- Compiled and executed the program by entering the ls command.
- The Child PID was displayed, followed by the execution of the given command, while the parent process waited for the child process to finish.

```
samhi@Samhita:~$ nano practical_1.c
samhi@Samhita:~$ gcc practical_1.c
samhi@Samhita:~$ ./a.out
ls
ls
Child PID is = 546
 a.out      exec1.c     f1       fork.c        linux_task4     permission_lab   pipetask1.c     student_portal
 abc.c      exec2.c     f1.c     fork1         new             pipe             pipetask2.c     task1.c
 copy       exec3.c     f11      fork3.c       newfile         pipe.c           practical_1.c
 copy.c     exec4.c     f2       fork3.c.save  newfile1        pipe3.c          practical_1.v
 copy.txt   exec5.c     f3       honey.c       open2.c         pipe4.c          practical_3
 def        exectask.c  filter  'honey^2'      os_practicals   pipeex.c         sample.txt
Parent PID is = 545
```

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigated the relationship between hardware resources and operating system services

```
samhi@Samhita:~$ uname
Linux
samhi@Samhita:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   16
  On-line CPU(s) list:    0-15
Vendor ID:                GenuineIntel
  Model name:             13th Gen Intel(R) Core(TM) i7-13620H
    CPU family:           6
    Model:                186
    Thread(s) per core:   2
    Core(s) per socket:   8
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             5836.80
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                          e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable
                           nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2api
                          c movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowpre
                          fetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi
                          1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv
                          1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_cl
                          ear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
```

```
Caches (sum of all):
  L1d:                     384 KiB (8 instances)
  L1i:                     256 KiB (8 instances)
  L2:                      10 MiB (8 instances)
  L3:                      24 MiB (1 instance)
NUMA:
  NUMA node(s):            1
  NUMA node0 CPU(s):       0-15
Vulnerabilities:
  Gather data sampling:    Not affected
  Ghostwrite:              Not affected
  Indirect target selection: Not affected
  Itlb multihit:           Not affected
  L1tf:                    Not affected
  Mds:                     Not affected
  Meltdown:                Not affected
  Mmio stale data:         Not affected
  Old microcode:           Not affected
  Reg file data sampling:  Mitigation; Clear Register File
  Retbleed:                Mitigation; Enhanced IBRS
  Spec rstack overflow:    Not affected
  Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_
                           IS_S
  Srbds:                   Not affected
  Tsa:                     Not affected
  Tsx async abort:         Not affected
  Vmscape:                 Not affected
```

```
samhi@Samhita:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                               /
samhi@Samhita:~$ ps
    PID TTY          TIME CMD
    336 pts/0    00:00:00 bash
    596 pts/0    00:00:00 ps
samhi@Samhita:~$ top
top - 17:03:04 up 51 min,  1 user,  load average: 0.05, 0.06, 0.06
Tasks:  27 total,   1 running,  26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7755.6 total,   7091.9 free,    590.7 used,    225.4 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7165.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    606 samhi     20   0    8152   5996   3912 R   0.3   0.1   0:00.01 top
      1 root      20   0   24188  15396  11584 S   0.0   0.2   0:00.99 systemd
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     46 root      19  -1   42208  16628  15348 S   0.0   0.2   0:00.23 systemd-journal
     81 systemd+  20   0   22416  14492  12000 S   0.0   0.2   0:00.09 systemd-resolve
     88 root      20   0   35472  12352   9160 S   0.0   0.2   0:00.33 systemd-udevd
    157 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.04 chronyd-starter
    158 root      20   0    4472   3168   2924 S   0.0   0.0   0:00.00 cron
    159 message+  20   0    8800   5396   4732 S   0.0   0.1   0:00.05 dbus-daemon
```

uname: Displays basic information about the operating system and system kernel.

lscpu: Displays detailed information about the system’s CPU.

lsblk: Displays information about the available block devices and storage drives.

ps: Displays information about the currently running processes.

top: Displays real-time information about system processes and resource usage.